

What is a Lead Time and Why Does it Matter?

A lead time is the delay between placing an order and receiving it. If a warehouse orders 50 units from its supplier on Monday and the lead time is 3 days, those units arrive on Thursday. Until Thursday, the warehouse cannot use them.

This sounds simple but it changes the inventory model in a fundamental way. With lead times, you have to order in advance of when you need the stock. If you do not plan ahead, goods arrive too late and you run out.

With a lead time of L periods, what you order in period t does not arrive until period $t + L$. So what arrives in period t is whatever was ordered in period t minus L :

$$I_w[t] = I_w[t-1] + o_w[t - L] - o_r[t]$$

With lead time L : what arrives now was ordered L periods ago

Going from One Period to Many

Once you have a time index t , the natural extension is to run the model across T periods instead of just one. Each period the warehouse receives stock from the supplier, ships to the retailer, and the retailer sells to customers. Costs accumulate across all periods and the solver minimises the total.

With $T = 5$ and 4 variables per period (o_w, o_r, I_w, I_r), the LP now has 20 decision variables instead of 4. Writing them all out manually is not practical. This is why we need dictionary variables and `lpSum`.

lpSum and Dictionary Variables

```
o_w = pulp.LpVariable.dicts('o_w', range(T), lowBound=0)
```

```
# sum across all T periods using lpSum
m += pulp.lpSum(
    c_order * o_w[t] + h_w * I_w[t] + h_r * I_r[t]
    for t in range(T)
)
```

Writing Constraints in a Loop

```
# One loop writes all T balance constraints
for t in range(T):
    # Handle lead time: what arrives in period t was ordered t-LT periods ago
    if t - LT >= 0:
        received = o_w[t - LT]    # order placed LT periods ago arrives now
    else:
        received = 0              # nothing ordered before period 0

    # Previous period inventory
    prev_Iw = I_w[t-1] if t > 0 else I_w0
    prev_Ir = I_r[t-1] if t > 0 else I_r0

    # Warehouse balance constraint for period t
    m += I_w[t] == prev_Iw + received - o_r[t]

    # Retailer balance constraint for period t
    m += I_r[t] == prev_Ir + o_r[t] - demand[t]
```

What to Watch

Video	What it covers
PuLP Basics — Lists and Dictionaries Channel: DataCamp youtube.com/watch?v=qNA7Y57bXLQ	Covers exactly how LpVariable.dicts works and how to access variables using keys. \
Using IpSum in PuLP Channel: DataCamp youtube.com/watch?v=SIAMHsb6jLU	Dedicated video on IpSum

Deliverable

A complete code for this setup.

Then try changing:

Lead time

T

And other things to experiment.